

For determination of open-circuit fault, the far end of the cable, in case of single cable, is earthed (or the far ends of the cables are inter-connected), an ammeter is inserted in series with the cable and low voltage supply is connected across the cable and earth or cables. Zero deflection in ammeter indicates open-circuit fault.

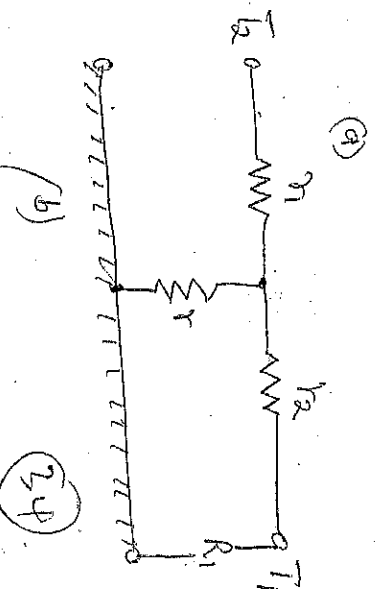
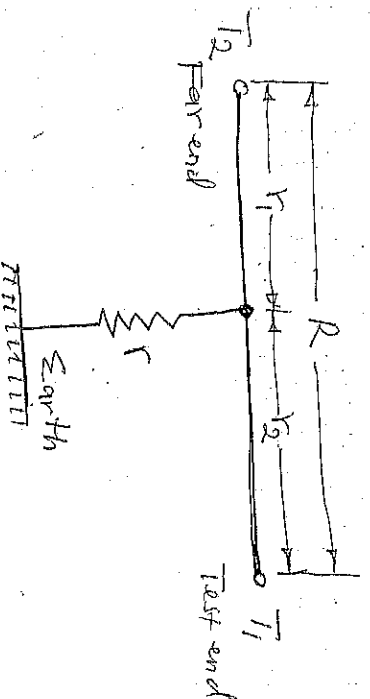
LOCATION OF CABLE FAULTS

There are various methods for locating the point of fault

1 - B Lavier 5 1287

Blavier's test is used to locate the ground fault of a single cable i.e. when no other cables run along with the faulty one.

This test is performed with the aid of a low voltage supply and either an ammeter or voltmeter with a bridge network.



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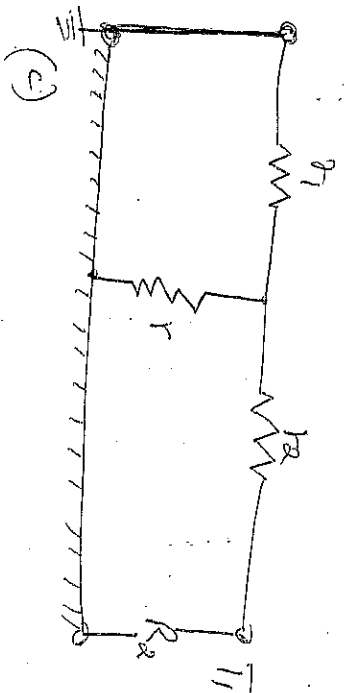


Figure 1: Blavier's test for Ground Fault of a single wire cable.

In this test resistance between one end of the cable r_1 and earth is measured first with the far end r_2 isolated from earth and then with the far end r_2 earthed as shown in Figure 1(c). Let the two readings be R_1 and R_2 respectively.

If r_1 and r_2 are the conductor resistances of the lengths of cable 'far end' to fault and 'test end' to fault respectively and r is the resistance of fault to earth then

$$R_1 = r_2 + r$$

--- (1)

$$\text{and } R_2 = \frac{r_1 r_2}{r_1 + r} + r_2$$

--- (2)

The total resistance of the conductor is
 $R = r_1 + r_2$

Substituting $r = R_1 - r_2$ from eqn. (1) in equation (2) we have

$$R_2 = r_2 + \frac{(R_1 - r_2) r_1}{(R_1 - r_2) + r_1}$$

$$r_2^2 R_2 - \sqrt{(R_1 - R_2)(R_1 - R_2)}$$

--- (4)

(35)

If the total length of the cable is L meters, the length of the cable between far end and fault is L_1 meters, length of cable between test end and fault is L_2 meters and cross-section of conductor is uniform then

$$\frac{L_1}{L_2} = \frac{r_1}{r_2}$$

Add 1 to both sides

$$\text{and } \frac{L_1 + L_2}{L_2} = \frac{r_1 + r_2}{r_2}, \text{ but } L_1 + L_2 = L$$

$$\text{or then } \frac{L}{L_2} = \frac{r_1 + r_2}{r_2}$$

$$L_2 = L \times \frac{r_2}{r_1 + r_2}$$

(5)

Thus the distance of fault from test end can be determined.

Murray Loop Tests

These tests are performed for the location of either an earth fault or a short-circuit fault in underground cables provided that a sound cable runs along with the grounded or a short-circuited cable. In these tests the resistance of fault does not affect the results obtained except when the resistance of fault is very high. There are two loop tests usually used and are known as Murray loop and Varley loop tests. These tests employed the principle of Wheatstone bridge.

1. Murray Loop Test

The connection diagrams to locate earth fault and short circuit fault by Murray loop test method are shown in Figures 2 and 3 respectively. Wheatstone bridge principle is used in these tests.

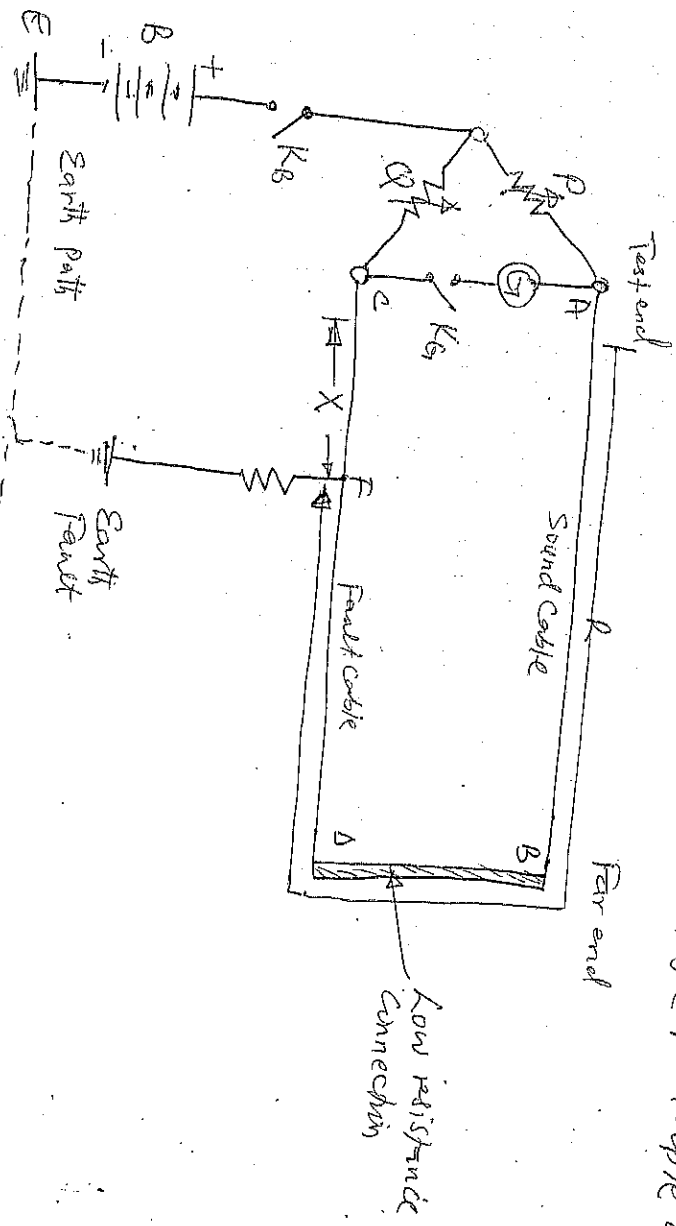


Figure 2: Murray loop test (Earth fault)
AB is the Sound Cable and CD is the faulty cable; the earth fault occurring at point F

(37)

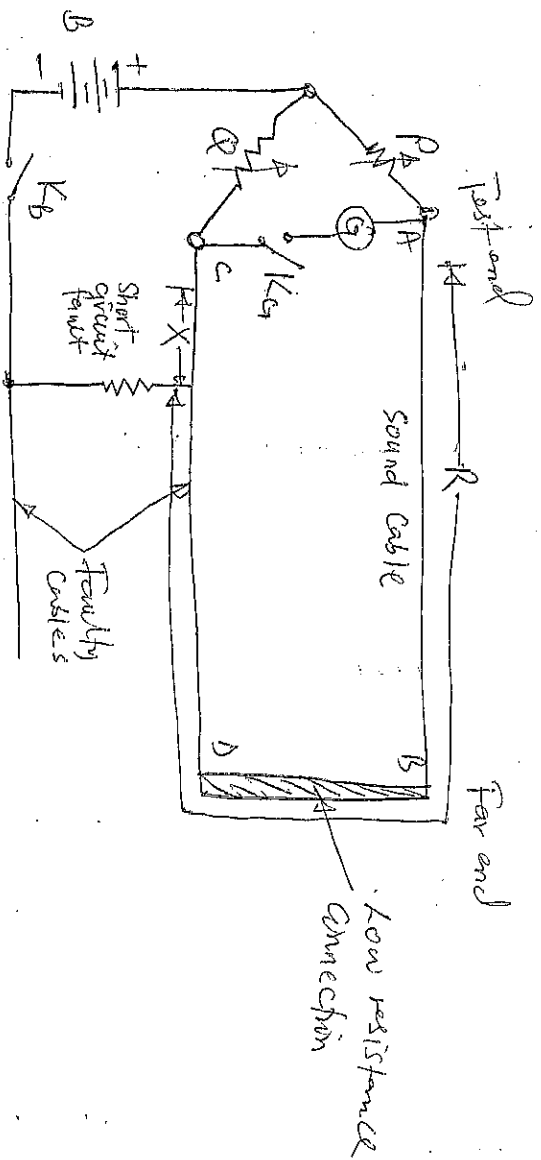


Figure 3: Murray loop test (short circuit fault).

P and Q are the two ratio arms consisting of step resistors and slide wire, G is the galvanometer, B is the battery, K_b is a battery key and K_s is a galvanometer key. The fault cable is connected to a second cable through low resistance link at the far end. Bridge is balance by adjusting the resistance of ratio arms P and Q until the galvanometer indicates zero. Let $R =$ resistance of the conductors up to the fault from the test end. While $x =$ resistance of the other length of the loop.

$$\frac{P}{Q} = \frac{R}{x}$$

Add 1 to both sides

$$\text{or } \frac{P}{Q} + 1 = \frac{R}{x} + 1$$

If r is the resistance of each cable, then $R + x = 2r$

$$\text{or } \frac{P}{Q} + 1 = \frac{R + x}{x} = \frac{2r}{x}$$

Where $2r =$ total resistance of one of the cables.

$$\text{or } x = \frac{Q}{P + Q} \times 2r$$

(38)

where r is resistance of one short wire cables when free from fault. If l is the length of each cable in meters, then

The resistance per meter length of cable = r/l

~~The distance of the fault from the testing end is~~

~~Distance of the fault from the testing end is~~
~~Distance per meter length of cable~~

c: Distance of fault from testing end ~~is~~ $d = \frac{X}{r/l} = \frac{lX}{r} = \frac{Q}{P+Q} \times 2r \frac{l}{r}$

$= \frac{Q}{P+Q} \times 2l \text{ meters} \dots (6)$

or $= \frac{Q}{P+Q} \times (\text{loop length}) \text{ meters.}$

The connection for short-circuit test in figure 3 are practically the same as in earth fault test except that a portion of one of the short-circuiting cables is used instead of earth as return path in the battery circuit. Balance is obtained as before.

At balance $\frac{P}{Q} = \frac{R}{X}$, Add 1 to both sides gives $\frac{P+Q}{Q} = \frac{R+X}{X}$

or $\frac{Q}{P+Q} = \frac{X}{R+X} \quad (\text{inverted})$

or $X = \frac{Q}{P+Q} (R+X) \dots (7)$

where $(R+X)$ is the total resistance of the loop and is assumed to be known.

Since P , Q and the total resistance of the loop is known so X and hence the distance of fault from line end of ratio arm Q can be determined.

VARLEY LOOP TEST

In this test also Sound Cable is required in addition to the existing cable. The circuit diagrams to determine the location of ground fault and short-circuit fault by method of Varley loop tests are shown in Figure 4 and Figure 5 respectively.

The difference between Murray loop and Varley loop tests is that in Varley loop test provision is made for the measurement of total loop resistance instead of obtaining it from the relation $R = \rho \frac{l}{a}$.

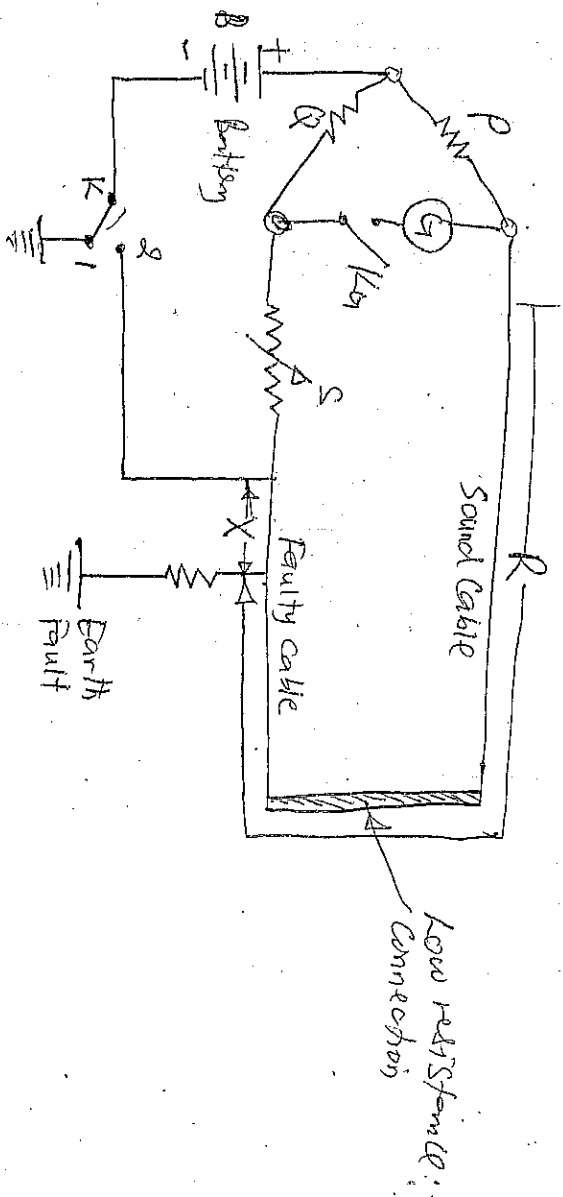


Figure 4: Varley loop test (Earth Fault)

In this test the ratio arms P and Q are fixed and balance is obtained by varying the known variable resistances. Balance is obtained once the key in the battery circuit on Stud 1 and then on Stud 2. At balance for earth fault or short-circuit fault with key K on Stud 1.

$$\frac{P}{Q} = \frac{R}{X+S_1}$$

where S_1 is the setting of variable resistance S .

$$\text{or } \frac{P+Q}{Q} = \frac{R+X+S_1}{X+S_1} \quad \text{add 1 to both sides}$$

$$\text{or } X = \frac{Q(R+X) - PS_1}{P+Q} \quad \dots (8)$$

At balance for earth fault or short-circuit fault
When key K on stud 2,

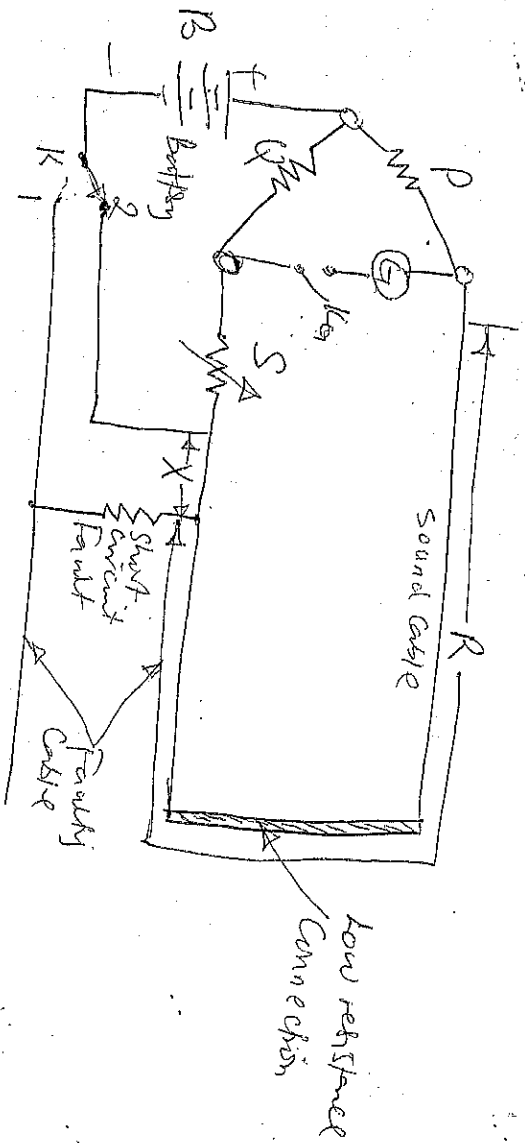


Figure 5: Varley loop test (Short-circuit)

$$\frac{P}{Q} = \frac{R+X}{S_2}$$

where S_2 is the new setting of variable resistance S .

$$\text{or } (R+X)Q = PS_2$$

$$\text{or loop resistance } (R+X) = \frac{PS_2}{Q}$$

(4)

(9)

From equations (8) and (9) we have:

$$X = \frac{P(S_2 - S_1)}{P + Q}$$

(10)

Thus X can be determined from the known values of P, Q, S_2 and S_1 .

Example 1.

In locating a ground fault by Varley loop test the following observations were recorded:

(i) With key K at Stud 1, a balance was obtained when $P = 10$, $Q = 80$ and $S = 2,400 \Omega$.

(ii) With key K at Stud 2, a balance was obtained

when $P = 10$, $Q = 80$ and $S = 3,400 \Omega$.

Determine the distance of the fault from the testing end. Each conductor is 5 km long.

Soln: -

$$P = 10 \Omega; Q = 80 \Omega; S_2 = 3,400 \Omega; S_1 = 2,400 \Omega$$

$$\therefore X = \frac{P(S_2 - S_1)}{P + Q} = \frac{10(3,400 - 2,400)}{10 + 80} = 111.11 \Omega$$

$$\text{Loop resistance, } P + X = \frac{P}{Q} S_2 = \frac{10}{80} \times 3,400$$

$$\text{or } r = 425 \Omega$$

$$\text{Resistance per km length of cable} = \frac{r}{L} = \frac{425}{10} = 42.5 \Omega$$

$$\text{Distance of fault from testing end} = \frac{X}{42.5} = \frac{111.11}{42.5} = 2.623 \text{ km}$$

(42)

Palmer

Example 2

In a test by Murray loop for ground fault on 1,000 mtrs of cable having a resistance of 1.6Ω per km, the faulty cable is looped with a sound cable of the same length and connection. If the ratio of the two arms of testing network at balance are in the ratio of 3:1, Calculate the distance of the fault from the testing end of the cable.

Soln: -

The ratio of two arms, $\frac{P}{Q} = 3$

$$\Rightarrow \frac{P+Q}{Q} = 3 + 1 = 4$$

Total resistance of loop, $(R+X) \times 2$

$$\left[\begin{array}{l} \frac{P}{Q} + 1 = \frac{3}{1} + 1 \\ \frac{P+Q}{Q} = 3 + 1 \end{array} \right] \text{From method 1 on both side}$$

$$\therefore R+X = \frac{1.6}{1000} \times 2 \times 1000$$

$$= 3.2$$

$$= 3.2 \Omega$$

$$\text{From eqn } \oplus \quad X = \frac{Q}{P+Q} (R+X) = \frac{1}{4} \times 3.2 = 0.8 \Omega$$

Distance of fault from testing end is given by

$$\frac{X}{R} = \frac{0.8}{1.6/1000} = 500 \text{ meters}$$

(43)

Example 3-

Murray loop test is performed on a faulty cable 300m long. At balance, the resistance connected to the faulty core was set at 15Ω and the resistance of the resistor connected to the sound core was 45Ω. Calculate the distance of the fault point from the test end.

Soln: -

Here, $Q = 15\Omega$; $P = 45\Omega$; loop length = $2 \times 300 = 600m$

$$d = \frac{Q}{P+Q} \times \text{loop length}$$

$$\therefore d = \frac{15}{45+15} \times 600 = 150m$$

Example 4.

In a test by Murray loop for ground fault on 520m of cable having a resistance of 1.6 Ω/km, the faulty cable is looped with a sound cable of the same length and area of cross-section. If the ratio of the other two arms of the loop network at balance is 3:1. Find the distance of the fault from the test end of cables.

Soln: -

$$\text{Here, } \frac{P}{Q} = \frac{3}{1} \quad \text{or} \quad \frac{P+Q}{Q} = 3+1$$

$$\text{or } \frac{P+Q}{Q} = 4$$

Distance of fault from test end is

$$d = \frac{Q}{P+Q} \times \text{loop length}$$

L

$$Q = 15$$

$$P = 45$$

$$\frac{Q}{P+Q}$$

(48)

Example 3

$$d = \frac{1}{4} \times (2 \times 500) = 250 \text{ m}$$

Example ~~49~~

In a test for a fault to earth on a 500m length of cable having a resistance of 1 Ω per 1000m, the faulty cable is looped with a sound cable of the same length but having a resistance of 2.25 Ω per 1000m. The resistance of the other two arms of the testing network at balance are in the ratio 2.75:1. Calculate the distance of the fault from the testing end of the cable.

Soln:-

$$\frac{P}{Q} = 2.75 \quad \text{or} \quad \frac{P+Q}{Q} = 2.75 + 1 = 3.75$$

$$\text{Resistance of loop} = \frac{1}{1000} \times 500 + \frac{2.25}{1000} \times 500 = 1.625 \Omega$$

Resistance of faulty cable from test end up to fault point is

$$X = \frac{Q}{P+Q} \times (\text{loop resistance}) = \frac{1}{3.75} \times 1.625 = 0.433 \Omega$$

Distance of fault point from the testing end is

$$d = \frac{X}{1/1000} = \frac{0.433}{1/1000} = 0.433 \times 1000 = 433 \text{ m}$$

(49)

Example 10

Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single healthy conductor is 20 Ω . The loop is completed with a similar conductor connected to the faulty conductor is 200 Ω . The fixed resistors have equal values. Calculate the distance of the fault from the test end.

Soln:-

$$\text{Here } P = Q; S = 200 \Omega; \therefore R =$$

Resistance of faulty cable from test end to fault point

$$X = \frac{Q(R + X) - P^2}{P + Q} \quad \text{from eqn (8)}$$

$$R + X = 20(20 + 20) = 800 \Omega$$

$$X = \frac{Q(800) - (Q \times 200)}{Q + Q} = 300 \Omega$$

The resistance per km = 20 Ω

$\therefore$ Distance of the fault from test end is

$$d = \frac{X}{r/l} = \frac{300}{20} = 15 \text{ km}$$

$r/l = 20 \Omega/\text{km}$ given

TRANSducers

(1)

INTRODUCTION

An electronic instrumentation system consists of a number of components which together are used to perform a measurement and record the result. An instrumentation system generally consists of three major elements: an input device, a signal conditioning or processing device, and an output device. The input device receives the quantity under measurement and converts a proportional electrical signal to the signal-conditioning device. Here the signal is amplified, filtered, or otherwise modified to a format acceptable to the output device. The output device may be a simple indicating meter, an oscilloscope, or a chart recorder for visual display. It may be a magnetic tape recorder for temporary or permanent storage of the input data, or it may be a digital computer for data manipulation or process control. The kind of system depends on what is to be measured and how the measurement result is to be presented.

The input quantity for most instrumentation systems is non-electrical. In order to use electrical methods and techniques for measurement, manipulation or control, the non-electrical quantity is converted into an electrical signal by a device called a transducer. Thus transducer is defined as: a device which, when actuated by energy in one transmission system, supplies energy in the same form or in another form to a second transmission system. The energy transmission may be electrical, mechanical, chemical, optical (radiant), or thermal.

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TYPES OF ELECTRIC TRANSDUCERS

Various types of electrical transducers are as follows:

- (i) Resistive type (ii) Capacitive type (iii) Inductive type.

RESISTIVE TYPE TRANSDUCER

In such a transducer, resistance between the output terminals of a transducer gets varied according to the measured. Resistive transducers are prepared over other transducers because dc and ac both are suitable for resistance measurement.

Resistance of any metal conductor is given by the equation

$$R = \rho \frac{L}{a}$$

... (1)

Where ρ is the resistivity of the material of conductor in $\Omega \cdot m$, L is the length of conductor in meters, and a is the cross-sectional area of conductor in m^2 .

Physical phenomenon of input signal to the transducer causes variation in resistance by changing any one of the quantities, ρ , L and a . For measurement of displacement length of conductor is varied in potentiometer thereby resulting in change in resistance. For measurement of force and pressure resistance of a conductor or semiconductor is varied in strain gauge. Variation in temperature causes change in the resistivity of the conductor material and so change in resistance takes place which is used for measurement of temperature. With some devices resistance varies with the change in light intensity because of photoconductive effect, while with others resistance varies on exposure to magnetic field due to magnetoresistive effect. Thus resistive transducers have a wide range 100 Ω - 100 k Ω .

A.

(46)

2. Transducers operating on the principle of variation of ⁽⁵⁾ Mutual inductance.

Such transducers operate on the fact that mutual inductance between two coils depends upon the self inductance of the coils and coefficient of coupling between them, as mutual inductance between two coils is given by the expression

$$M = K \sqrt{L_1 L_2}$$

--- (3)

where K is the coefficient of coupling and L_1 and L_2 are self inductances of the coils.

In such transducers two or more coils are used and the mutual inductance between the coils is varied as per requirement.

A simple application of inductive transducer using this principle for measurement of linear displacement is shown in Figure 6. In such a transducer, both the coils are in series so have inductance of both the coils as an algebraic sum of self inductance of both coils and mutual inductance between the coils.

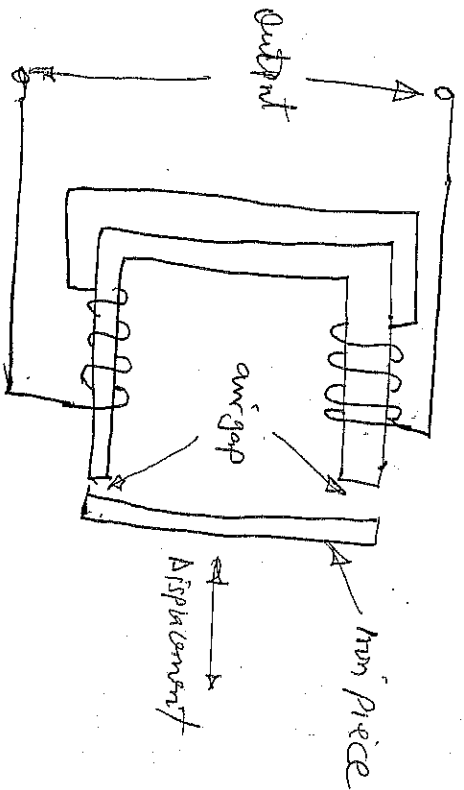


Figure 6: Inductive transducer for measurement of linear displacement

5. Transducers operating on the principle of production of ⁽⁷⁾ eddy currents.

These transducers operate on the principle that when a conducting plate is kept near a coil carrying alternating current, eddy currents are induced in the conducting plate, producing its own magnetic field in opposition to the main field created by the coil. Thus eddy currents induced in the conducting plate reduce the net flux linking with the coil and so the inductance of the coil is reduced. The nearer is the plate to the coil, the higher are the induced eddy currents and so higher is the reduction in the inductance of the coil. Thus, the inductance of the coil changes with the movement of the plate.

An inductive transducer based on this principle is shown in Figure 8.6.

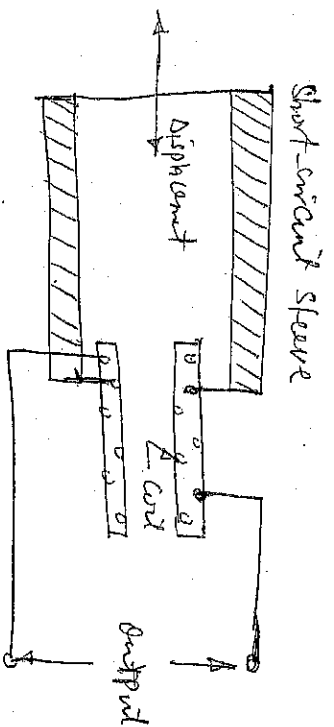


Figure 8.6: Inductive transducer for measurement of linear displacement

In such a transducer there is a coil carrying an alternating current and a cylindrical sleeve of some ferromagnetic material moving parallel and coaxially to the coil. When the sleeve covers less length of coil, low eddy currents are induced in it so more is the inductance of coil and when sleeve covers more length of the coil, more eddy currents are induced in it and so less is the inductance of the coil. This change in inductance, the measure of displacement, is sensed and indicated by an

① *

- (1) By Varying overlapping area of Plates, A .
- (ii) By Varying distance between Plates, d .
- (iii) By Varying relative permittivity of dielectric material between two Plates.

The above changes can be caused by physical variable like displacement, force or pressure. The change in capacitance may also be caused by change in permittivity as in the case of measurement of levels of liquids or gases.

The change in capacitance can be measured with an a.c. bridge, but it is usually measured with an oscillatory circuit. The transducer, as a part of the oscillatory circuit, causes a change in the frequency of the oscillator. This change in frequency is a measure of the magnitude of the non-electrical quantity under measurement.

Variable Capacitance transducers find considerable usage in specific and limited areas in the field of displacement measurement. They are ideally suited as non-contact type dynamic sensors, especially, in the studies of vibration in very light structures such as thin walls and diaphragms. The non-contact type sensors are desirable when a transducer is required for measurement of mechanical displacement without causing any additional mechanical loading on the vibrating object.

LINEAR VARIABLE DIFFERENTIAL TRANSFORMER (LVDT)

This is the most widely used inductive transducer for translating linear motion into an electrical signal. The basic construction of an LVDT is shown in Figure below.

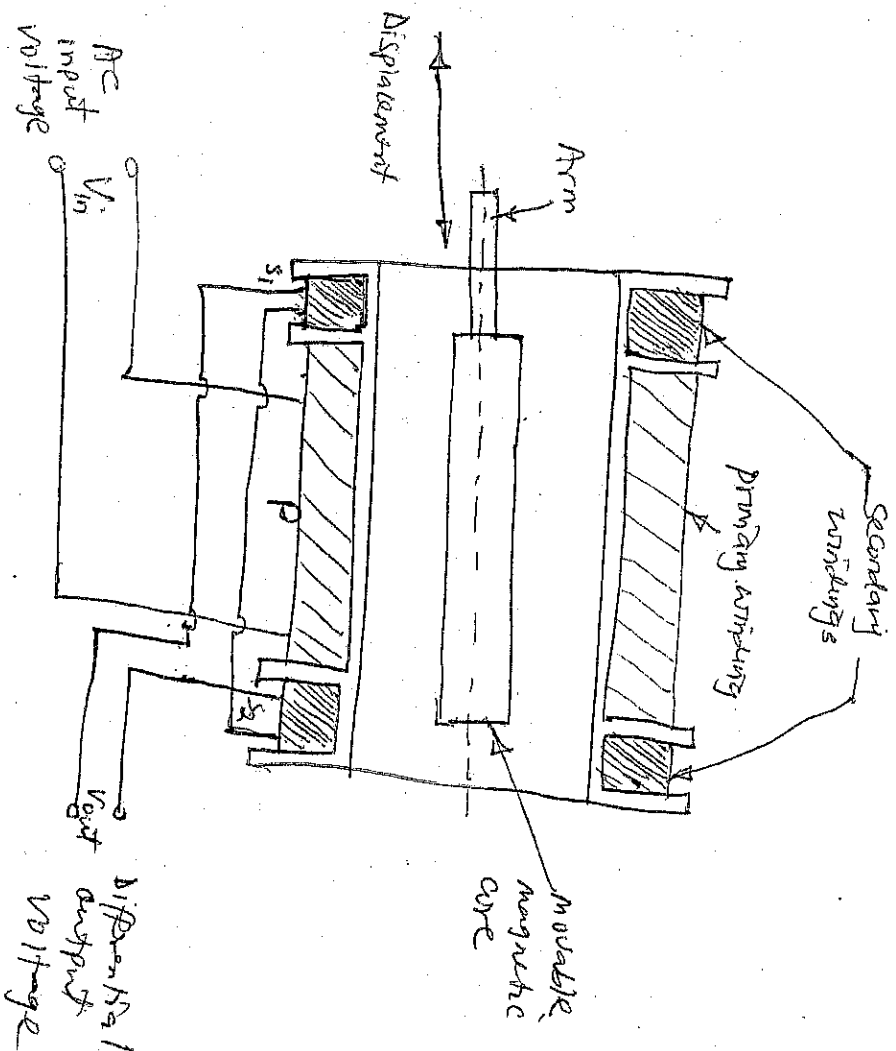
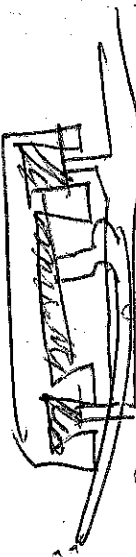


Figure LVDT

LVDT is a differential transformer consisting of one primary winding P and two identical secondary windings S_1 and S_2 , wound over a hollow bobbin of non-magnetic and insulating material. The secondary windings S_1 and S_2 , which have equal number of turns are arranged concentrically and placed either side of the primary winding P . A soft iron core, attracted to the sensing element of which displacement is to be measured, in the shape of rod or cylinder slides freely in the hollow



by measuring the magnitude and comparing the phase of differential output voltage with the input primary voltage, amount and direction of displacement of core may be determined.

Example

The output of an LVDT is 1.25V at maximum displacement. At a load of 0.75mN the deviation from the linearity is maximum and it is ± 0.0025 V from a straight line through origin. Determine the linearity at a given load.

Soln:-

Maximum deviation from linearity $\Delta_{\max} = 0.0025$ V
Output Voltage, $V_{\text{out}} = 1.25$ V

$$\text{Line } \frac{P_{\max}}{P_{\text{out}}} \text{ Also } \frac{V_{\max}}{V_{\text{out}}}$$

$$\text{Linearity at the given load} = \frac{\Delta_{\max}}{V_{\text{out}}} \times 100 = \frac{0.0025}{1.25} \times 100 = 0.2\%$$

Example 10

An LVDT is used for measuring the deflection of a bellows. The sensitivity of LVDT is 40V/mm. The bellows is deflected by 0.125mm by a pressure of 0.8×10^6 N/m². Determine the sensitivity of the LVDT in V per mm and the pressure when the voltage output of LVDT is 3.5V.

Soln:-

LVDT Sensitivity, $S = 40$ V/mm

Output voltage for a deflection of 0.125mm, $V_{\text{out}} = 40 \times 0.125 = 5$ V

$$\text{Hence the sensitivity of LVDT} = \frac{V_{\text{out}}}{P} = \frac{5}{0.8 \times 10^6}$$

$$S = \frac{V_{\text{out}}}{P} = 6.25 \times 10^{-6} \text{ V per N/m}^2$$

(13)

$$\frac{V_{\text{out}}}{S} = \frac{5}{8}$$

$$P = \frac{V_{\text{out}}}{S}$$

$$S = \frac{V_{\text{out}}}{P}$$

It can also be shaped ~~can~~ ^{be} of a magnetic material ^{be} used as the core. The input shaft fastened to the core (rotor) is mounted at the center of the coil former on which the primary and secondary windings are wound symmetrically. The can also be shaped as (rotor) is so chosen as to give linear output over a specific angle of rotation. The shaft is mounted with miniature precision ball bearings so that the friction and mechanical hysteresis is well reduced to minimum.

In null position of the core i.e. in central position, output voltage induced in secondary windings S_1 and S_2 are equal and in ~~opposite~~ ^{opposite} therefore the differential output is zero. Angular displacement of the rotor from the null position gives differential voltage output which is proportional to the angular displacement.

Closewise rotation of the core produces increasing differential output voltage of one phase while counter-clockwise rotation produces increasing differential output voltage of opposite phase. So by measuring the magnitude and comparing the phase of differential output voltage with the input primary voltage the amount of angular displacement and its direction may be determined.

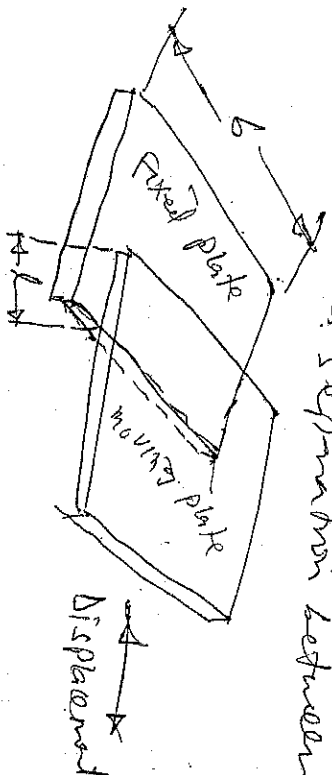
$$S = \frac{V_{out}}{P}$$
$$\frac{E_{max}}{V_{out}} \times 100.$$

CAPACITIVE TRANSDUCERS - By variation of overlapping area of plates.

Such a transducer operates on the fact that any capacitor is proportional to the overlapping area of plates. Consider a capacitive transducer having parallel plates of constant width b , as shown in Figure below. In such a transducer, capacitance is proportional to l where l is the length of overlapping portion of the plates and if varies change in displacement under measurement. So any for a parallel plate capacitor, the capacitance C is given by an expression:

$$C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{\epsilon_0 \epsilon_r b l}{d} \text{ Farads}$$

where l is the length of overlapping portion, b is the width of plate in meter and d is separation between plates in meter.



Figure

Capacitive transducer by variation of overlapping area of plates

where l is the length

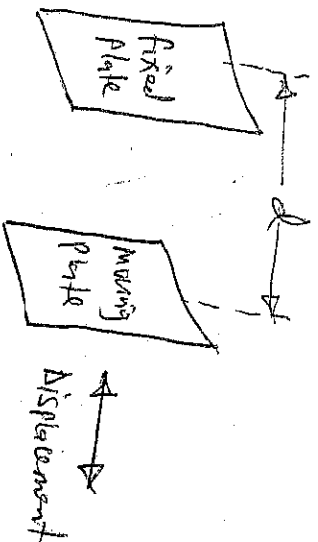
Sensitivity, $S = \frac{\delta C}{\delta l} = \frac{\epsilon_0 \epsilon_r b}{d} \text{ F/m}$

(16)

$$Q = \frac{Q}{P+Q}(2l) = \dots$$

capacitive transducers - by variation of distance between plates.

This type of transducer operates upon the fact that the capacitance of any capacitor is inversely proportional to the distance between the plates. Such a capacitive transducer is used for measurement of linear displacement as shown in figure below.



Figure

This transducer consists of one plate fixed and other moving. As the moving plate moves away from or towards the fixed plate as its displacement under measurement, the capacitance decreases or increases. This change in capacitance is measured by an ac bridge circuit or with an oscillating circuit and thus displacement of the moving plate is determined.

From the expression for capacitance of a parallel plate capacitor, small change in capacitance due to small displacement is given as

$$\frac{\delta C}{C} = \frac{-\epsilon_0 \epsilon_r A}{C d^2}$$

From the above expression it is obvious that the sensitivity, being equal to $\frac{\delta C}{C}$ or $\frac{-\epsilon_r A}{d^2}$ is not constant but varies over the transducer range.

Example

(iv)

A capacitance transducer of two parallel plates of overlapping area of $5 \times 10^{-4} \text{ m}^2$ is immersed in water. The capacitance has been found to be 9.5 pF . Calculate the separation d between the plates and the sensitivity, $S = \frac{\delta C}{\delta d}$ of the transducer. Given ϵ_r for water = 81; $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

Soln: -

$$\text{Since } C = \frac{\epsilon_0 \epsilon_r A}{d}$$

$$\text{Separation between plates, } d = \frac{\epsilon_0 \epsilon_r A}{C} = \frac{8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{9.5 \times 10^{-12}}$$

$$= 37.75 \times 10^{-3} \text{ m or } 37.75 \text{ mm}$$

$$\text{Sensitivity, } S = \frac{\delta C}{\delta d} = \frac{-\epsilon_0 \epsilon_r A}{d^2} = \frac{-8.854 \times 10^{-12} \times 81 \times 5 \times 10^{-4}}{(37.75 \times 10^{-3})^2}$$

$$= 0.025 \times 10^{-8} \text{ F/m}$$

Example

A capacitive transducer is made up of two concentric cylindrical electrodes. The length of electrodes is 25 mm , the inner diameter of outer cylindrical electrode is 4.8 mm and the outer diameter of inner cylindrical electrode is 4 mm . Determine the sensitivity of the transducer. Assume air medium.

Determine the change in capacitance for a displacement (20)

The strain gage is an example of a passive transducer that converts a mechanical displacement into a change of resistance and further converts it to electrical signals. A strain gage is a thin, wafer-like device that can be attached (bonded) to a variety of materials to measure applied strain. Metallic strain gages are manufactured from small diameter resistance wire, such as constantan or etched from thin foil sheets. The resistance of the wire or metal foil changes with length as the material to which the gage is attached undergoes tension or compression. This change in resistance is proportional to the applied strain and is measured with specially adapted Wheatstone bridge.

The sensitivity of a strain gage is described in terms of a characteristic called the gage factor, K , defined as the unit change in resistance per unit change in length. The gage factor can be written as

$$G = \frac{\Delta R/R}{\Delta L/L}$$

$$= 1 + 2\nu + \frac{\Delta P/P}{\Delta L/L} \quad \dots (5)$$

Change in resistance
due to change in length
Change in resistance
due to change in area
Change in resistance
due to piezoresistive effect.

Thus the strain $\frac{\Delta L}{L}$ can be measured in terms of gage factor, G and $\Delta R/R$

The strain $\frac{\Delta L}{L} = \epsilon$ and is usually measured in microstrain

($\mu \text{ strain/m}$)

$\frac{\Delta P/P}{\Delta L/L}$ can be expressed in terms of π where π is called the longitudinal π coefficient and ϵ is called the young's modulus of elasticity. π can be either +ve or -ve

Check on 3rd page.

(2)

Example

A resistance strain gauge with a gauge factor of 2 is fastened to a steel rod, subjected to a stress of 500 kg/cm^2 . If the modulus of elasticity of the steel is $2 \times 10^6 \text{ kg/cm}^2$, calculate the change in resistance of the strain gauge element due to the applied stress.

Soln:

Steel stress, $P = 500 \text{ kg/cm}^2$ Young modulus of elasticity of steel, $E = 2 \times 10^6 \text{ kg/cm}^2$

$$\text{Strain, } \epsilon = \frac{P}{E} = \frac{500}{2 \times 10^6} = 0.00025$$

Change in resistance, $\frac{\Delta R}{R} = \epsilon \times G$

$$= 0.00025 \times 2$$

$$= 0.0005 \text{ or } 0.05\%$$

$$\epsilon = \frac{P}{E} \quad \text{Gauge factor } G = \frac{\Delta R/R}{\epsilon}$$

$$P = \frac{\Delta R}{G} \times \frac{R}{\epsilon}$$

(26)

$$\frac{\Delta R}{R} = \frac{P}{E} \times G$$

TOP TYPES OF ELECTRICAL RESISTANCE STRAIN GAUGES

③

Based on mounting, strain gauges are classified as: Bonded strain gauges and unbonded strain gauges.

1. Bonded strain gauge - Bonded strain gauges are attached to the elastic element surface. They are primarily used for strain analysis. In bonded resistance wire strain gauges resistance element is bonded to the base which may be a thin sheet of paper or a thin sheet of bakelite or Teflon. The bonded strain gauge is connected to the Wheatstone bridge circuit.

The wire is covered on top with a thin sheet of material in order to protect it from any mechanical damage. The spreading of wires allows a uniform distribution of stress over the grid. Two connecting leads are soldered or welded to the ends of the grid. The bonded strain gauge is cemented with a special adhesive to the structure whose tensile or compressive strain is to be measured.

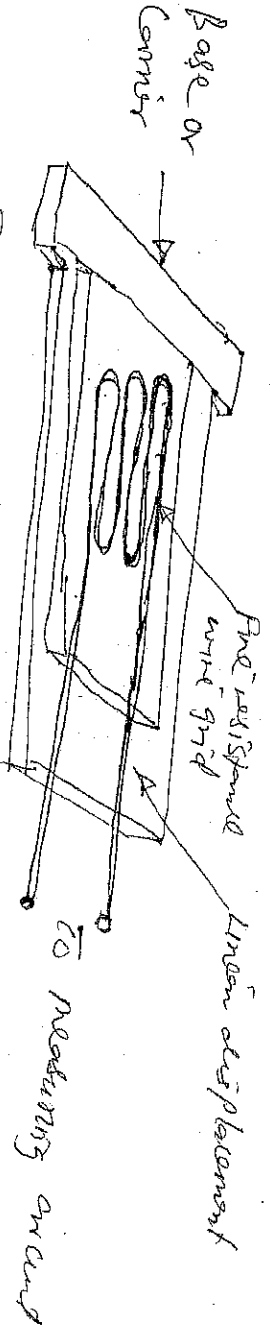


Fig. Bonded wire strain gauge

WIRE STRAIN GAUGES

The resistance element is in the form of wire foil or film of the material.

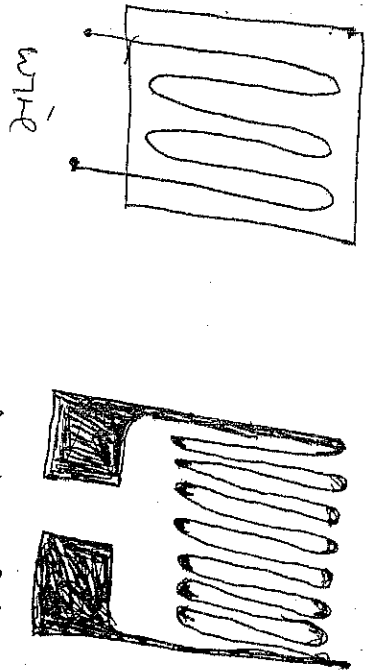


Figure strain gauge element.

Wire strain gauge - The resistance element is in the form of wire foil or film of the material.

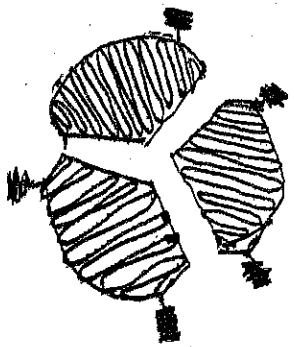
Metal foil - The strain is detected by using a metal foil as shown in figure (b). The metal foil is pasted on one side of plastic. The leads are soldered to metal foil.

Connecting the Wheatstone bridge. Metal foil strain gauges exhibit a high gauge factor than wire foil strain gauges.

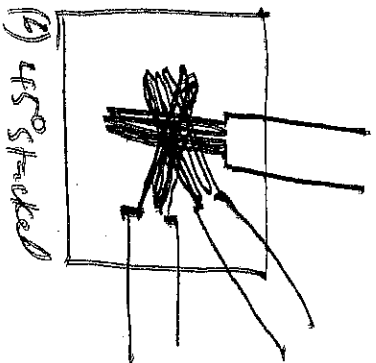
The metals and alloys used for the foil or wire are:
 Constantan (Cu-Ni), Nichrome (Ni-Cr), Isoelastic (Fe-Ni-Cr-Mn-Mo), Platinum-Tungsten (Pt-W).

ROSETTE STRAIN GAUGES

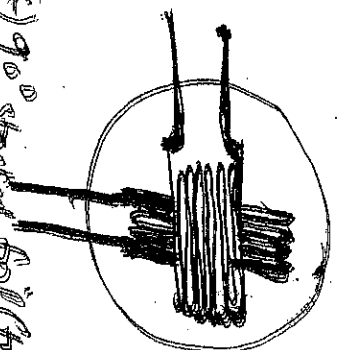
(5)



(a) 60° Planar foil
Three element rosette



(b) 45° stacked
line



(c) 90° stacked foil (two element)
strain gauge

Figure

Various types of rosette strain gauge

Strains that are parallel to the strain axis are indicated by a single strain gauge. whereas if there is a requirement of measurement of the strain in more than one direction, placing more than one strain gauge in a number of directions at the same location. Two element rosettes of foil gauges are commonly used in force transducers. The elements are mounted in a Wheatstone bridge circuit for providing maximum output. Three element rosettes are often used to determine the direction and magnitude of principal strains resulting from complex stress field loading.

OPERATING PRINCIPLE OF RESISTANCE STRAIN GAUGE

The working of strain gauge is based on the fact that when stress is applied on the metal conductor its resistance changes owing to change in length and cross-sectional area of the conductor as shown in figure. Resistance of conductor under stress is also changed due to change in resistivity. This property is called the piezo-resistive effect. That's why strain gauges are called the piezo-resistive strain gauges.

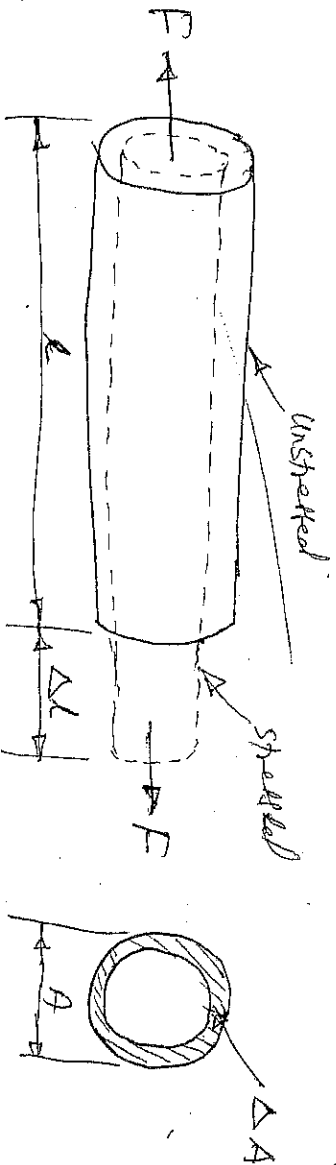


Figure Conductivity under axial tension

If a conductor of length L , area of cross-section A is subjected to axial tension, the resistance will change because of change in length ~~length~~ cross-sectional area ~~by ΔA~~ and resistivity ~~by Δρ~~ of the material.

Resistance of an unstrained conductor is given by expression

$$R = \rho \frac{L}{A}$$

... (1)

Let under strained condition's resistance of conductor be changed by ΔR because of change in length by ΔL , cross-sectional area by ΔA and resistivity by $\Delta \rho$. These quantities can be related with each other by differentiating equation (1) with respect to stress i.e.

$$\frac{dR}{d\rho} = \frac{d}{d\rho} \left(\rho \frac{L}{A} \right) = \frac{\rho}{A} \frac{dL}{d\rho} - \frac{\rho L}{A^2} \frac{dA}{d\rho} + \frac{1}{A} \frac{d\rho}{d\rho} \quad \dots (2)$$

(29)

The term $\Delta l/l$ in the denominator of equation (5) can be written as

$$G = \frac{\Delta R/R}{\delta} \quad (6)$$

Where δ = the strain in the lateral direction. The resistance change ΔR of a ~~free~~ conductor with length l of a conductor of uniform cross-section

$$R = \rho \frac{\text{length}}{\text{area}} = \frac{\rho \times l}{(\pi d^2/4)}$$

(7)

where ρ = the specific resistance of the conductor material
 l = length of the conductor
 d = diameter of the conductor

Tension on the conductor causes an increase Δl in its length and a simultaneous decrease in Δd in its diameter. The resistance of the conductor then changes to

$$R_s = \rho \frac{(l + \Delta l)}{(\pi/4)(d - \Delta d)^2} = \rho \frac{l(1 + \Delta l/l)}{(\pi/4)d^2(1 - 2\Delta d/d)} \quad (8)$$

Equation (8) may be simplified by using Poisson's ratio, μ , defined as the ratio of strain in the lateral direction to strain in axial direction. Therefore

$$\mu = \frac{\Delta d/d}{\Delta l/l} \quad \dots (9)$$

Thus it is deduced that

$$\frac{\Delta R/R}{\Delta l/l} = 1 + 2\mu + \frac{\Delta \rho/\rho}{\Delta l/l} = G, \quad \text{the gauge factor.}$$

(8)

Dividing both sides of equation (12) by R we get 14 by $R = \rho \frac{L}{A}$

$$\frac{1}{R} \frac{dR}{dP} = \frac{1}{L} \frac{dL}{dP} - \frac{1}{A} \frac{dA}{dP} + \frac{1}{\rho} \frac{d\rho}{dP}$$

--- (13)

For small variations, above expression can also be written as

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} - \frac{\Delta A}{A} + \frac{\Delta \rho}{\rho}$$

--- (14)

Under strained conditions, cross-sectional area of the conductor changes, so the diameter of the conductor is also changed.

$$\text{Area, } A = \frac{\pi D^2}{4}$$

$$\text{or } \frac{\delta A}{\delta S} = 2 \times \frac{\pi}{4} \frac{\delta D}{\delta S}$$

Dividing both sides of above equation by A , we get

$$\frac{1}{A} \cdot \frac{\delta A}{\delta S} = \frac{2 \frac{\pi}{4} D}{\frac{\pi}{4} D^2} \cdot \frac{\delta D}{\delta S}$$

--- (15)

go to
page
2.

$$\frac{1}{A} \cdot \frac{\delta A}{\delta S} = \frac{2}{D} \frac{\delta D}{\delta S}$$

--- (16)

For small variations equation (16) can be written as

$$\frac{\Delta A}{A} = 2 \frac{\Delta D}{D}$$

--- (17)

(30)

Example

A strain gauge is cemented to an iron post of length 4m, which is subjected to a compressible force. The nominal resistance of the gauge is 200Ω. The change in resistance of strain gauge is measured to be 0.7Ω. The final length of the iron post is 999mm. Calculate the gauge factor of the strain gauge.

Soln: -

Initial length of wire, $L = 4m = 4000mm$

Final length of wire, $L' = 999mm$

Increase in length of wire, $\Delta L = L' - L = 999 - 4000 = -1mm$

$$\text{Strain, } \epsilon = \frac{\Delta L}{L} = \frac{-1}{4000} = -0.0001$$

Initial resistance of wire, $R = 200\Omega$

Change in resistance of wire, $\Delta R = -0.7\Omega$

∴ Under compression the resistance of the strain will decrease

$$\text{Gauge factor of the device, } G = \frac{\Delta R/R}{\epsilon} = \frac{-0.7/200}{-0.0001} = 3.5$$

* TYPES OF ELECTRICAL RESISTANCE STRAIN GAUGES

1. Wire strain gauge - wire strain gauges are normally of smaller size, and are subject to minimal leadage, and can be employed in high temperature applications.

Wire strain gauges are of two types namely

- (a) Unbonded wire strain gauge and
- (b) Bonded wire strain gauge.

(a) Unbonded wire strain gauges - In such type of gauge, resistance wire is not bonded on a base, instead of that strain is directly transferred to the resistance.

* With no external load applied it is at initial Preload, the strains and resistances of four wires are normally equal giving a balanced bridge i.e. output voltage $V_{out} = 0$. This is accomplished by providing adjustable resistances in the bridge.

Application of an external force increases tension in two wires and reduces it in other two wires causing unbalanced variation in resistance of wires unbalancing the bridge and causing an output voltage V_{out} in proportion to the displacement. The output device can be calibrated to read the magnitude of the displacement of the armature directly.

These devices are used mainly for measurement of force, pressure or strain.

(b) Bonded wire strain gauge - The most common type of bonded wire strain gauge used for detecting displacement caused by large forces is flat grid type as shown in Figure 10.

In flat grid type bonded wire strain gauge fine resistance wire of diameter of 25 micron or less is wound back and forth in a grid format with as many loops as possible, laid side by side. This grid is covered by base or carrier which may be a thin sheet of paper or bakelite or a sheet of teflon. The wire is covered on top with a thin sheet of material in order to protect it from any mechanical damage. The spreading of wires allows a uniform distribution of stress over the grid. A large length to width ratio in the grid structure is also desirable in order to keep the transverse sensitivity minimum. Two connecting leads are soldered or welded to the ends of the grid. The bonded strain gauge is cemented with a special adhesive to the structure whose tensile or compressive strain is to be measured.

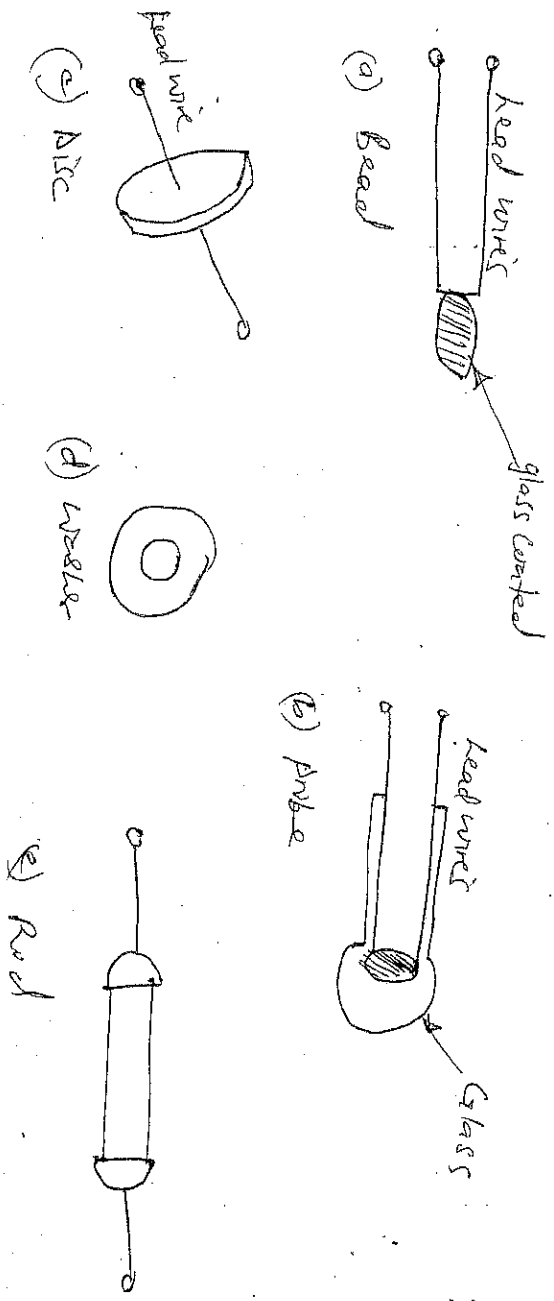


Figure 11: Commercial forms of thermistors.

THERMISTOR CHARACTERISTICS

Three important characteristics of thermistors noted especially in measurement and control applications are:

- (i) The resistance - temperature characteristics
- (ii) The voltage - current characteristics and
- (iii) The current - time characteristics

(i) The resistance - ~~character~~ - temperature characteristic shows that the thermistor has a very high negative temperature coefficient of resistance, making it an ideal temperature transducer.

(ii) The voltage - current characteristics shows that the voltage drop across a thermistor increases with increasing current until it reaches a peak value beyond which the voltage drop decreases as the current increases.

(iii) The current - time characteristics curve indicates the time delay to reach maximum current as a function of the applied voltage.

Thermocouples

Thermocouples are devices for measuring temperature.

A thermocouple consists of a pair of dissimilar metal wires joined together at one end (sensing, or hot junction) and terminated at the other end (reference, or cold junction) which is maintained at a known temperature (reference temperature). When a temperature difference exists between the sensing junction and the reference junction, an emf is produced that causes a current in the circuit,

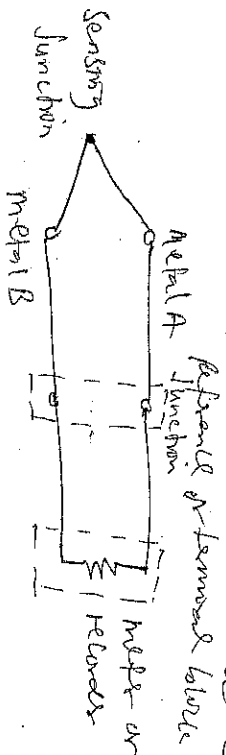


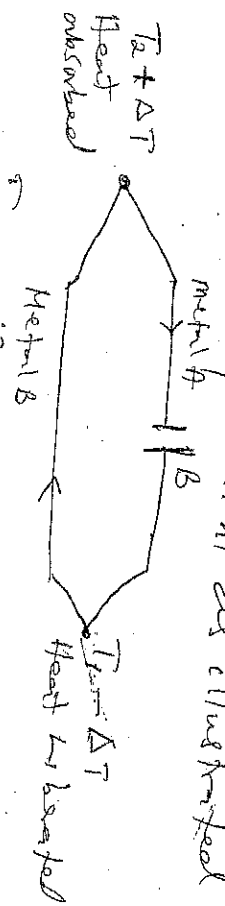
Figure 12: Basic thermocouple circuit

When the reference junction is terminated by a meter or recording instrument, as in Figure 12 above, the meter indication will be proportional to the temperature difference between the hot junction and the reference junction. This thermoelectric effect, caused by contact potentials at the junction, is known as the Seebeck effect, after the German Physicist Thomas Seebeck.

The magnitude of the thermal emf depends on the wire material used and on the temperature difference between the junctions.

The reverse of Seebeck effect is also true and that has been explained by Peltier effect. Peltier observed that when an electric current flows across a junction of two dissimilar metals

heat is either generated (liberated) or absorbed, depending upon the direction of flow of current as illustrated in Figure 13.



The relation between the thermocouple set up and temperature difference of hot and cold (or reference) junctions is given by the equation

$$E = \alpha(T - T_0) + \beta(T^2 - T_0^2) \quad \dots \dots \dots (1)$$

Where E is the thermoelectric e.m.f. set up in volts, T is the absolute temperature of hot junction, T_0 is the absolute temperature of cold or reference junction and α and β are constants whose values depend upon the metals used in the couple.

Example

A copper-constantan thermocouple has $\alpha = 37.5 \mu V/^\circ C$ and $\beta = 0.0045 \mu V/^\circ C$. Determine the e.m.f. developed by the thermocouple when its hot junction is at $200^\circ C$ and cold junction is kept in ice.

Soln. -

$$\text{Given } \alpha = 37.5 \mu V/^\circ C = 37.5 \times 10^{-6} V/^\circ C$$

$$\beta = 0.0045 \mu V/^\circ C = 0.0045 \times 10^{-6} V/^\circ C$$

Temperature of hot junction, $T = 200 + 273 = 473 K$

Temperature of cold junction, $T_0 = 0 + 273 = 273 K$

~~EMF~~ developed by the thermocouple,

$$E = \alpha(T - T_0) + \beta(T^2 - T_0^2)$$

UNUS OVERLAPPING THERMOCOUPLES
homogeneous wire is one that is physically and chemically the same throughout.

(14)

Law of homogeneous metals - This law states that a thermocouple circuit that is made with a homogeneous wire must generate an e.m.f., even if it is at different temperatures and the leads throughout. In other words, a thermocouple must be made from at least two different materials in order to generate voltage. A change in the area of the cross-section of wire, or a change in the temperature in different places in the wire, will not produce a voltage.

Law of intermediate metals - The sum of all of the e.m.f.'s in a thermocouple circuit using two or more different metals is zero if the circuit is at the same temperature. In another words the addition of different metals to a circuit will not affect the voltage the circuit creates if the added junctions are to be at the same temperature as the junctions in the circuit.

Law of Successive or Intermediate temperatures - A thermocouple made from two different metals produces an e.m.f. E_1 when the metals are at different temperatures T_1 and T_2 , respectively. Suppose one of the metals has a temperature change to T_3 , but the other remains at T_2 . Then the e.m.f. created when the thermocouple is at temperatures T_1 and T_3 will be the summation of the first and second, so that $E_{new} = E_1 + E_3$.

This law allows a thermocouple that is calibrated with a reference temperature to be used with another reference temperature. It also allows extra wires with the same thermoelectric characteristics to be added to the circuit without affecting its total e.m.f.